$Z_L = j\omega L$ $Z_C = \frac{1}{j\omega C}$

No.

Date.

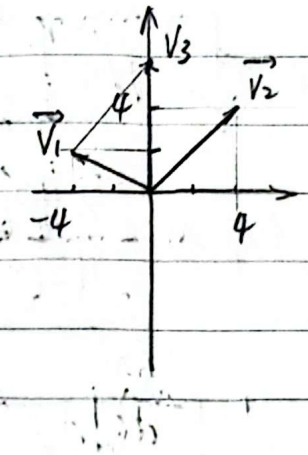
第四周作业

第五章

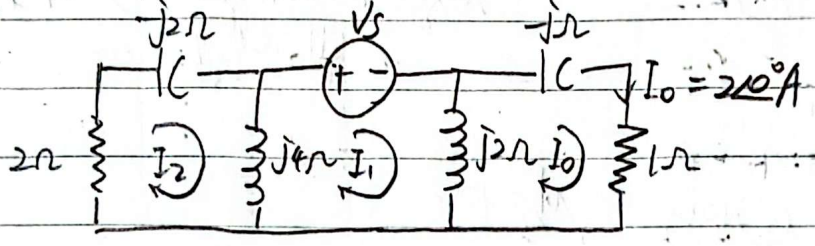
P1 T12 P2 T12、13 P3 T1、5、6、7、10、11、14、15、16

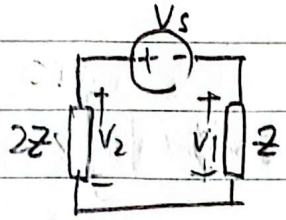
P1 T12 (1) $\vec{V_1} + \vec{V_2} = \vec{V_3} = 6\angle 90°\,V$

(2) $v_S = 10\sin(\omega t + \frac{\pi}{3}) = 10\cos(\omega t - \frac{\pi}{6}) = 10\angle -30°\,V$



P2 T12



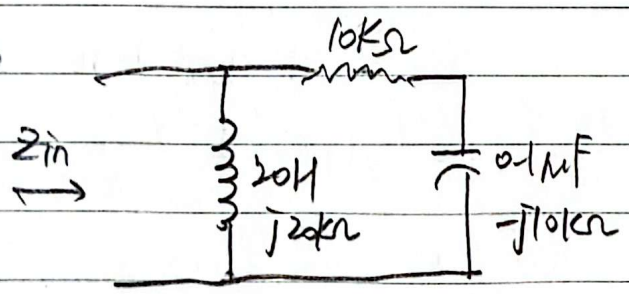


$V_1 = I_0(1-j) = 2(1-j)$

$V_2 = -2V_1 = -4(1-j)$

$V_2 = V_1 + V_S \Rightarrow V_S = V_2 - V_1 = -6(1-j) = 6\sqrt{2}\angle 135°$

P3 T13



$\omega = 1000\,rad/s$

$Z_R = 10k\Omega$ $Z_L = j\omega L = j20k\Omega$ $Z_C = \frac{1}{j\omega C} = -j10k\Omega$

$Z_{in} = (Z_R + Z_C) \| Z_L = 20k\Omega$

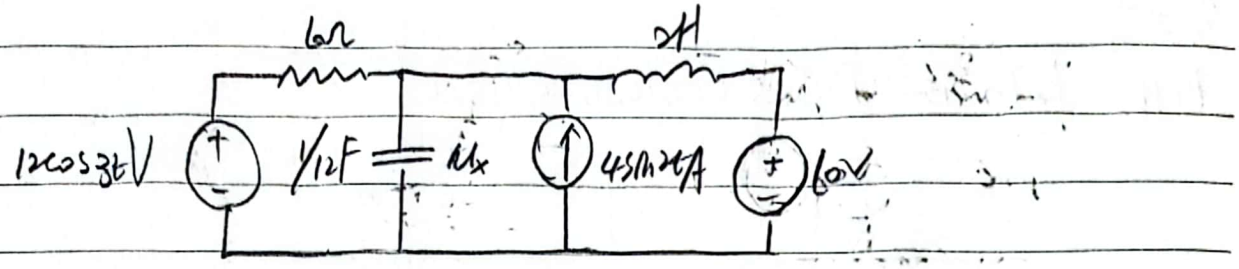
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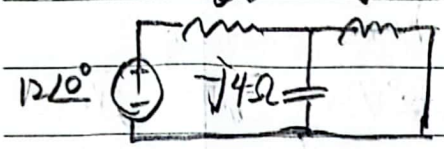
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$\frac{1}{j\omega C}$

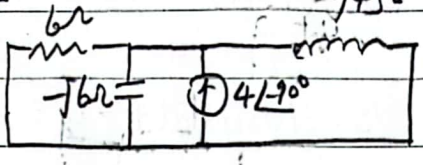
BT1



$\omega=3$

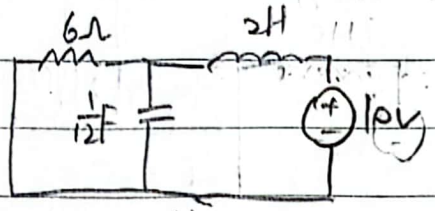


$\omega=2$



$u_{x1}=u\frac{z_c\|z_L}{z_c\|z_L+z_R}=10.7\angle-26.6°$

$u_{x2}=i(z_c\|z_L\|z_R)=21.5\angle-63.4°$



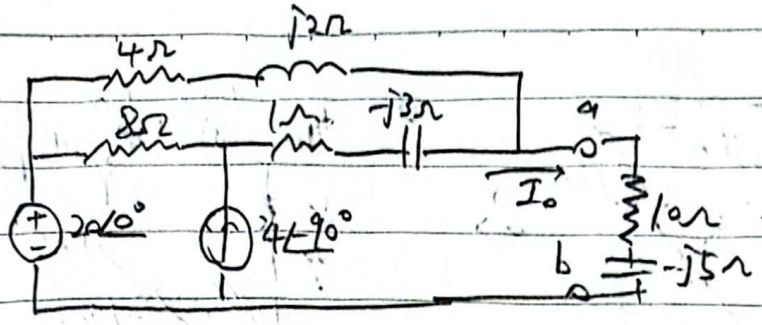
$u_{x3}=10V$

$u=u_{x1}+u_{x2}+u_{x3}=\left[10.7\cos(3t-26.6°)+21.5\cos(2t-63.4°)+10\right]V$

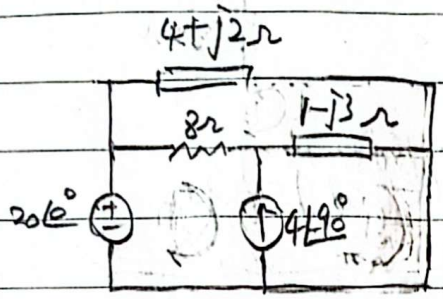
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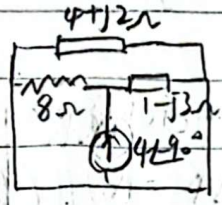
Date.

P3.5

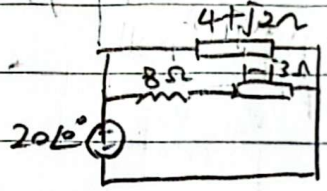


(1)



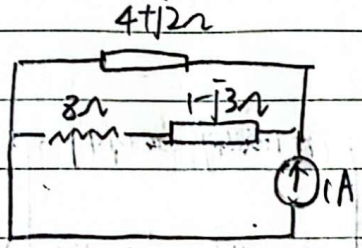


$$\Rightarrow i_1 = 4\angle -90° \frac{18}{8+1-j3} = \frac{16}{15} - \frac{16}{5}i$$



$$\Rightarrow i_2 = \frac{20\angle 0°}{(4+j2)\|(9-j3)} = 6 + \frac{4}{3}i$$

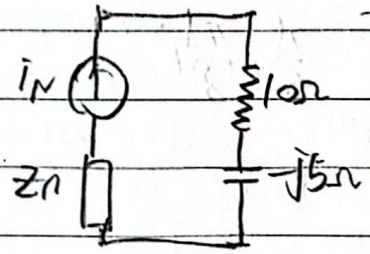
$$i_N = i_1 + i_2 = \frac{106}{15} - \frac{68}{15}i = 8.40\angle -32.68° \text{ A}$$



$$V = (8+1-j3)\|(4+j2) = \frac{54}{17} + \frac{12}{17}i \text{ V}$$

$$Z_N = \frac{54}{17} + \frac{12}{17}i \ \Omega$$

(2)



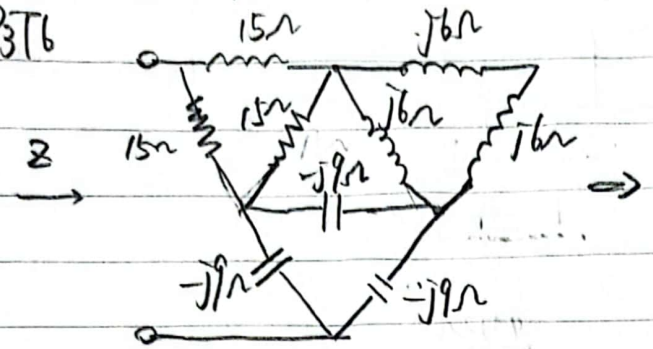
$$I_o = I_N = \frac{106}{15} - \frac{68}{15}i$$

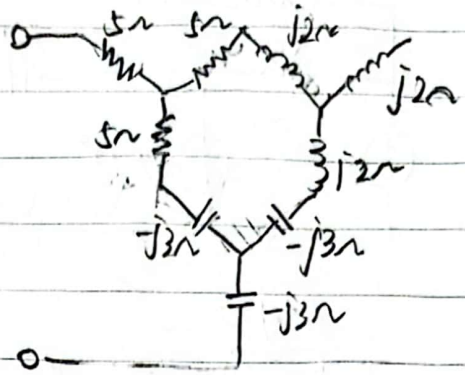
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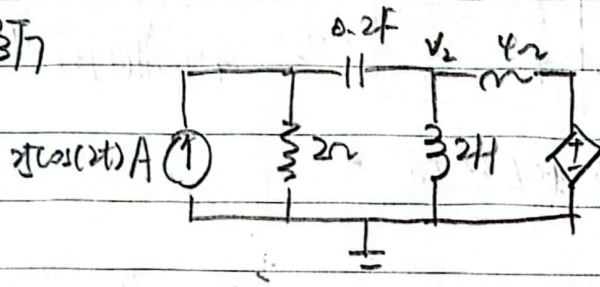
P3T6



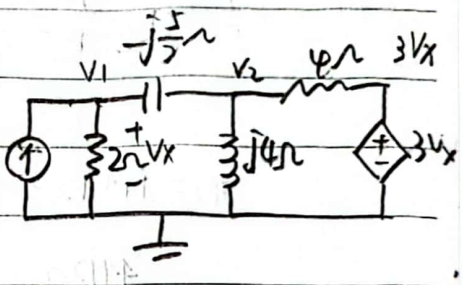


$$Z = 5-j3+(5-j3)\,\|\,(5+j3) = \left(\frac{205}{26}-j\frac{39}{26}\right)\Omega$$

P3T7



$\omega=2$



$$\begin{cases} \frac{1}{5}V_1 + 0.4j(V_1-V_2) = 2\angle 0° \\ \frac{1}{4j}V_2 + \frac{1}{4}(V_2-3V_x) + 0.4j(V_2-V_1) = 0 \end{cases} \Rightarrow \begin{bmatrix} \frac{1}{5}+0.4j & -0.4j \\ -\frac{3}{4}-0.4j & \frac{1}{4}+0.15j \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2\angle 0° \\ 0 \end{bmatrix}$$

$$\Rightarrow \left(\frac{5}{16}+0.5625j\right)V_1 = -\frac{25}{8}+\frac{125}{8}j$$

$$\Rightarrow V_1 = 28.32\angle 60.02°$$

$$\Rightarrow V_2 = 82.56\angle 57.13°$$

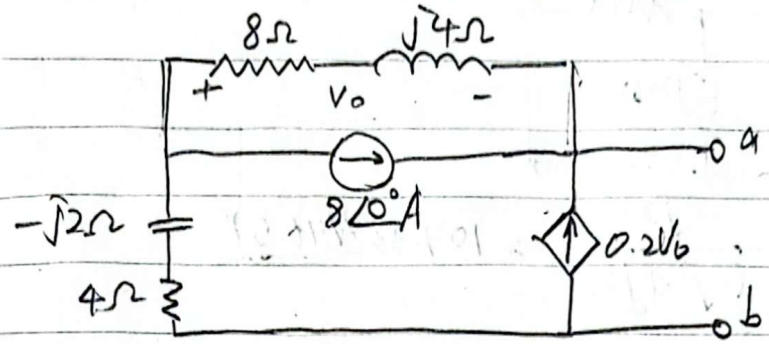
故 $\begin{cases} v_1(t) = 28.32\cos(2t+60.02°) \\ v_2(t) = 82.56\cos(2t+57.13°) \end{cases}$

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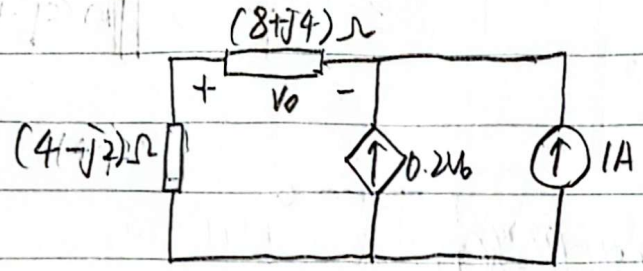
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P3T10



$(0.2V_o+8)(8+j4) = -V_o \Rightarrow V_o = \frac{-64-32j}{2.6+0.8j}$

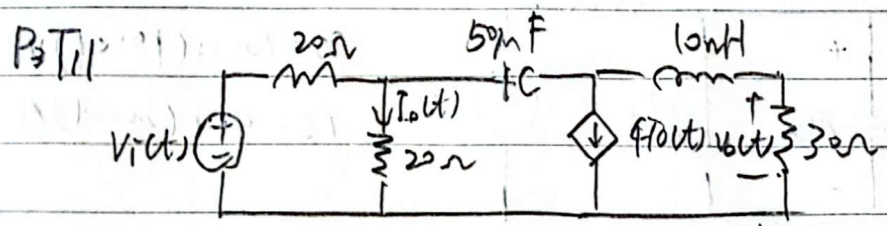
$U_{ab} = -V_o + 0.2V_o(4-j2) = (-0.2-0.4j)V_o = 11.76\angle 72.90°$



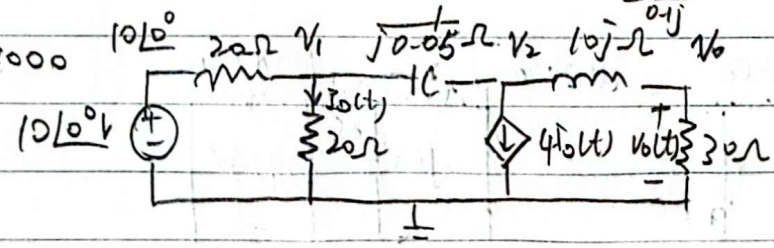
$(1+0.2V_o)(8+j4) = -V_o \Rightarrow V_o = \frac{-8-4j}{2.6+0.8j}$

$R = \frac{V}{I} = (0.2V_o+1)(12+2j) = 4.47\angle -7.64°\ \Omega$

P3T11



$\omega = 1000$



故 $v(t) = 6.15\cos(1000t + 0.00°)$

$\frac{1}{20}(V_1-10) + \frac{1}{20}V_1 + 0.05j(V_1-V_2) = 0$

$0.05j(V_2-V_1) + 4\frac{1}{20}V_1 + \frac{1}{30+10j}V_2 = 0$

$\Rightarrow \begin{cases} V_1 = 1.057\angle -24.15° \\ V_2 = 6.49\angle 88.41° \end{cases}$

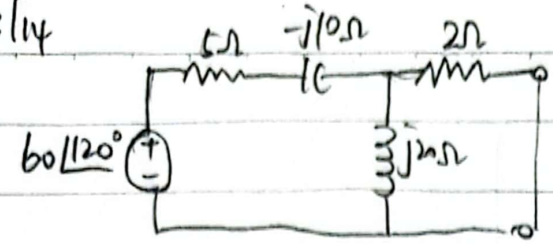
$V_o = V_2 \frac{30}{10j+30} = 6.15\angle 70.00°$

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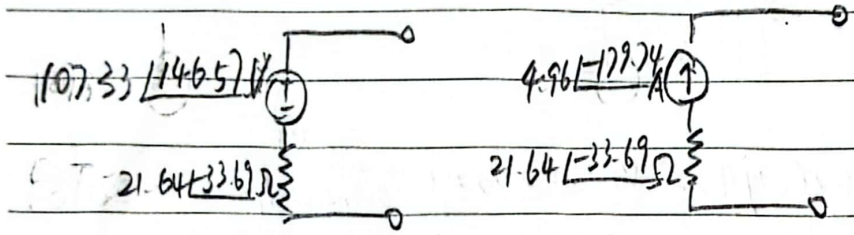
P3T14



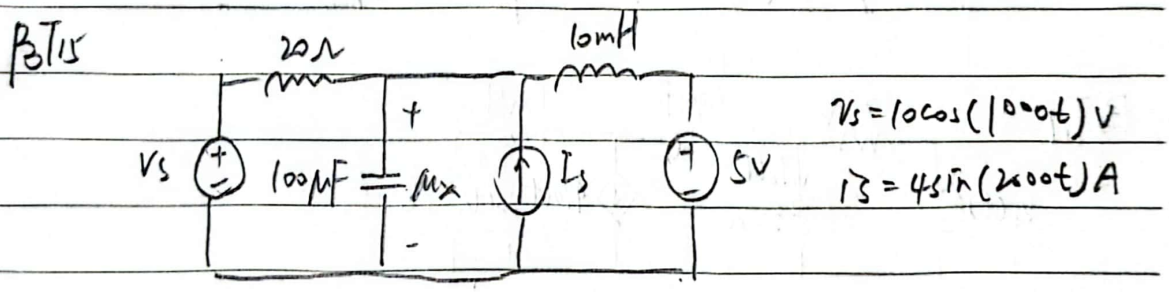
$$U_o = 60\angle 120^\circ \cdot \frac{j20}{5-j10+j20} = 107.33\angle 146.5^\circ$$

$$I = \frac{60\angle 120^\circ}{\frac{5-j10+2\|j20}{2}} = 4.96\angle -179.74^\circ$$

$$R = \frac{U}{I} = 21.64\angle -33.69^\circ$$

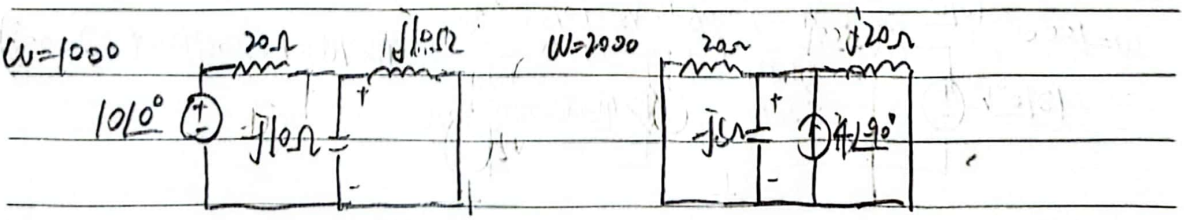


P3T15

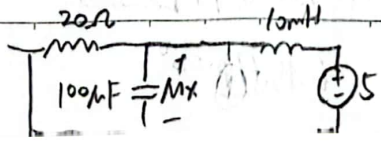


$v_s = 10\cos(1000t)\,V$

$i_s = 4\sin(2000t)\,A$



$$U_1 = 10\angle 0^\circ \times \frac{(-j10)\|j10}{20+j10\|(-j10)} = 10\angle 0^\circ \qquad U_2 = 4\angle -90^\circ(-j5\|j20\|20) = 8\sqrt{10}\angle -161.5^\circ$$



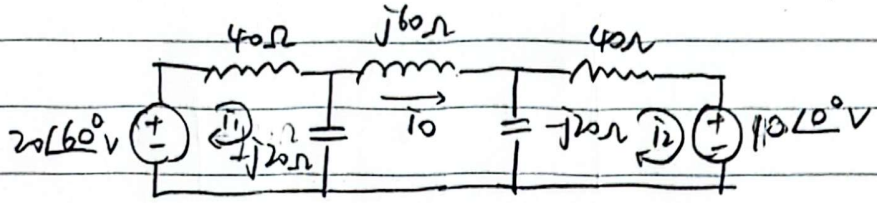
$U_3 = 5V$

$U = 10\cos(1000t + 8\sqrt{10}\cos(2000t - 161.5^\circ) + 5$

No.

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P3T16



$$(40-j20)i_1+j20i_0=20\angle 60°$$

$$(j60-j20-j20)i_0+j20i_1+j20i_2=0$$

$$(40-j20)i_2+10\angle 0°+j20i_0=0$$

$$\Rightarrow \begin{bmatrix} 20j & 40-20j & 0 \\ 20j & 20j & 20j \\ 20j & 0 & 40-20j \end{bmatrix} \begin{bmatrix} i_0 \\ i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 20\angle 60° \\ 0 \\ -10 \end{bmatrix}$$

$$\Rightarrow i_0 = 0.24\angle -33.69°$$